

Formalism

Quantum Mechanics

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1 Linear Algebra

Linear algebra provides the mathematical language for quantum mechanics. Key concepts include vector spaces, inner products, linear transformations, and eigenvalue problems. Scalars are generally complex.

1.1 Vectors

A vector space consists of vectors $(|\alpha\rangle, |\beta\rangle, \dots)$ and scalars $(a, b, \dots)$ with:

- **Vector addition:** Commutative, associative, null vector $|0\rangle$, inverse $|- \alpha\rangle$.
- **Scalar multiplication:** Distributive, associative, $0|\alpha\rangle = |0\rangle$, $1|\alpha\rangle = |\alpha\rangle$.

A **linear combination** is $a|\alpha\rangle + b|\beta\rangle + \dots$. A set of vectors is **linearly independent** if none can be written as a combination of the others. A set that spans the space and is linearly independent is a **basis**. The number of basis vectors is the **dimension**.

With respect to a basis $\{|e_i\rangle\}$, any vector is uniquely represented by components:

$$|\alpha\rangle = \sum_{i=1}^n a_i |e_i\rangle \leftrightarrow (a_1, a_2, \dots, a_n).$$

1.2 Inner Products

The inner product $\langle\alpha|\beta\rangle$ is a complex number satisfying:

$$\langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*, \quad (1)$$

$$\langle\alpha|\alpha\rangle \geq 0, \quad \langle\alpha|\alpha\rangle = 0 \Leftrightarrow |\alpha\rangle = |0\rangle, \quad (2)$$

$$\langle\alpha|(b|\beta\rangle + c|\gamma\rangle) = b\langle\alpha|\beta\rangle + c\langle\alpha|\gamma\rangle. \quad (3)$$

A vector space with an inner product is an **inner product space**. The **norm** is $\|\alpha\| = \sqrt{\langle\alpha|\alpha\rangle}$. Vectors are **orthogonal** if $\langle\alpha|\beta\rangle = 0$. An **orthonormal** set satisfies $\langle\alpha_i|\alpha_j\rangle = \delta_{ij}$.

For an orthonormal basis:

$$\langle\alpha|\beta\rangle = \sum_{i=1}^n a_i^* b_i, \quad \langle\alpha|\alpha\rangle = \sum |a_i|^2, \quad a_i = \langle e_i|\alpha\rangle.$$

The **Schwarz inequality**:

$$|\langle\alpha|\beta\rangle|^2 \leq \langle\alpha|\alpha\rangle \langle\beta|\beta\rangle.$$

1.3 Linear Transformations

A linear transformation $\hat{T}$ satisfies:

$$\hat{T}(a|\alpha\rangle + b|\beta\rangle) = a\hat{T}|\alpha\rangle + b\hat{T}|\beta\rangle.$$

With respect to a basis:

$$\hat{T}|e_j\rangle = \sum_{i=1}^n T_{ij} |e_i\rangle,$$

where T_{ij} are matrix elements. In an orthonormal basis, $T_{ij} = \langle e_i|\hat{T}|e_j\rangle$. The transformation is represented by a matrix $\mathbf{T}$. For a vector $|\alpha\rangle$ with components $\mathbf{a}$, $\hat{T}|\alpha\rangle$ has components $\mathbf{a}' = \mathbf{T}\mathbf{a}$.

1.3.1 Matrix Operations and Terminology

- **Transpose:** $\tilde{T}_{ij} = T_{ji}$.
- **Complex conjugate:** $(\mathbf{T}^*)_{ij} = T_{ij}^*$.
- **Hermitian conjugate (adjoint):** $\mathbf{T}^\dagger = \tilde{\mathbf{T}}^*$.
- **Commutator:** $[\mathbf{S}, \mathbf{T}] = \mathbf{ST} - \mathbf{TS}$.
- **Trace:** $\text{Tr}(\mathbf{T}) = \sum_i T_{ii}$.
- **Determinant:** $\det(\mathbf{T})$ is the product of eigenvalues.
- **Inverse:** $\mathbf{T}^{-1}\mathbf{T} = \mathbf{T}\mathbf{T}^{-1} = \mathbf{1}$.
- **Hermitian:** $\mathbf{T}^\dagger = \mathbf{T}$.
- **Unitary:** $\mathbf{U}^\dagger = \mathbf{U}^{-1}$.

Under a change of basis $|\alpha^f\rangle = \mathbf{S}|\alpha^e\rangle$:

$$\mathbf{T}^f = \mathbf{S}\mathbf{T}^e\mathbf{S}^{-1}.$$

If both bases are orthonormal, $\mathbf{S}$ is unitary. Trace and determinant are invariant under similarity transformations.

1.4 Eigenvectors and Eigenvalues

An eigenvector satisfies:

$$\hat{T}|\alpha\rangle = \lambda|\alpha\rangle, \quad |\alpha\rangle \neq |0\rangle.$$

In matrix form: $\mathbf{T}\mathbf{a} = \lambda\mathbf{a}$, so $(\mathbf{T} - \lambda\mathbf{1})\mathbf{a} = 0$. Nonzero solutions exist iff:

$$\det(\mathbf{T} - \lambda\mathbf{1}) = 0.$$

This is the **characteristic equation**, an n th-order polynomial for λ . The eigenvectors can be chosen as a basis if they span the space. If so, $\mathbf{T}$ is diagonalizable:

$$\mathbf{T} = \mathbf{S}^{-1} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \mathbf{S}.$$

1.5 Hermitian Transformations

A Hermitian transformation satisfies $\hat{T}^\dagger = \hat{T}$, i.e., $\langle\alpha|\hat{T}|\beta\rangle = \langle\hat{T}\alpha|\beta\rangle$ for all $|\alpha\rangle, |\beta\rangle$.

Key properties:

1. Eigenvalues are **real**.
2. Eigenvectors belonging to distinct eigenvalues are **orthogonal**.
3. The eigenvectors **span the space** (for finite dimensions; infinite dimensions require careful treatment).

Thus, Hermitian operators can be diagonalized by a unitary transformation.

2 Function Spaces

Functions behave as vectors. The inner product is:

$$\langle f|g\rangle = \int f(x)^* g(x) dx.$$

The space of square-integrable functions, $L_2(a, b) = \{f : \int_a^b |f(x)|^2 dx < \infty\}$, is a Hilbert space.

2.1 Example: Legendre Polynomials

On $[-1, 1]$, polynomials of degree $< N$ form an N -dimensional space. Orthonormalizing the basis $\{1, x, x^2, \dots\}$ yields the Legendre polynomials:

$$P_0 = 1, P_1 = x, P_2 = \frac{1}{2}(3x^2 - 1), P_3 = \frac{1}{2}(5x^3 - 3x), \dots$$

They satisfy:

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}.$$

2.2 Operators in Function Spaces

Operators like $\hat{x}$ (multiplication by x) and $\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$ act on functions. Hermiticity requires boundary conditions. On $L_2(-\infty, \infty)$, $i\hat{D}$ is Hermitian, and $\hat{x}$ is Hermitian.

2.3 Continuous Spectrum and Dirac Delta

In infinite-dimensional spaces, Hermitian operators may have a continuous spectrum with eigenfunctions not in L_2 . These are normalized to delta functions:

$$\langle e_k | e_l \rangle = \delta(k - l).$$

For example:

- Position operator $\hat{x}$: eigenfunctions $e_{x'}(x) = \delta(x - x')$.
- Momentum operator $\hat{p}$: eigenfunctions $e_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$.

Completeness takes the form of an integral:

$$|\Psi\rangle = \int c_k |e_k\rangle dk, \quad c_k = \langle e_k | \Psi \rangle.$$

3 Generalized Statistical Interpretation

3.1 Postulates

1. The state of a particle is represented by a normalized vector $|\Psi\rangle$ in Hilbert space L_2 .
2. Observables $Q(x, p, t)$ are represented by Hermitian operators $\hat{Q}$. The expectation value is $\langle Q \rangle = \langle \Psi | \hat{Q} | \Psi \rangle$.
3. A measurement of Q yields one of the eigenvalues of $\hat{Q}$. The probability of obtaining eigenvalue λ is $|\langle e_\lambda | \Psi \rangle|^2$, where $|e_\lambda\rangle$ is the corresponding (normalized) eigenvector.

3.2 Discrete vs. Continuous Spectra

- **Discrete:** $|\Psi\rangle = \sum_n c_n |e_n\rangle$, $c_n = \langle e_n | \Psi \rangle$, probability $P(\lambda_n) = |c_n|^2$.
- **Continuous:** $|\Psi\rangle = \int c_k |e_k\rangle dk$, $c_k = \langle e_k | \Psi \rangle$, probability in dk is $|c_k|^2 dk$.

3.3 Position and Momentum Representations

- Position basis: $e_x(x') = \delta(x - x')$, $c_x = \Psi(x, t)$, $P(x)dx = |\Psi(x, t)|^2 dx$.
- Momentum basis: $e_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$, $c_p = \Phi(p, t)$ (momentum-space wave function), $P(p)dp = |\Phi(p, t)|^2 dp$.

These are Fourier transforms: $\Phi(p, t) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx/\hbar} \Psi(x, t) dx$.

4 The Uncertainty Principle

4.1 Generalized Uncertainty Principle

For any two observables A and B :

$$\sigma_A^2 \sigma_B^2 \geq \left(\frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right)^2. \quad (4)$$

This follows from the Schwarz inequality and the definition of standard deviation.

4.2 Position-Momentum Uncertainty

The canonical commutation relation:

$$[\hat{x}, \hat{p}] = i\hbar.$$

Therefore:

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}.$$

4.3 Minimum-Uncertainty Wave Packets

The equality $\sigma_x \sigma_p = \hbar/2$ is achieved when:

$$\left(\frac{\hbar}{i} \frac{d}{dx} - \langle p \rangle \right) \Psi = i\alpha(x - \langle x \rangle) \Psi,$$

with α real. The solution is a Gaussian:

$$\Psi(x) = A e^{-\alpha(x - \langle x \rangle)^2 / (2\hbar)} e^{i\langle p \rangle x / \hbar}.$$

4.4 Energy-Time Uncertainty Principle

For an observable Q with no explicit time dependence:

$$\frac{d\langle Q \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{Q}] \rangle.$$

Then from the generalized uncertainty principle with $A = H$, $B = Q$:

$$\sigma_H \sigma_Q \geq \frac{\hbar}{2} \left| \frac{d\langle Q \rangle}{dt} \right|.$$

Define $\Delta E = \sigma_H$ and $\Delta t = \sigma_Q / |d\langle Q \rangle / dt|$. Then:

$$\Delta E \Delta t \geq \frac{\hbar}{2}. \quad (5)$$

Here Δt is the time for $\langle Q \rangle$ to change by one standard deviation—a measure of how fast the system evolves. This is not a statement about measurement of time as a dynamical variable.

5 Key Results Summary

$$\text{Canonical commutation: } [\hat{x}, \hat{p}] = i\hbar. \quad (6)$$

$$\text{Generalized uncertainty: } \sigma_A \sigma_B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|. \quad (7)$$

$$\text{Position-momentum: } \sigma_x \sigma_p \geq \frac{\hbar}{2}. \quad (8)$$

$$\text{Energy-time: } \Delta E \Delta t \geq \frac{\hbar}{2}. \quad (9)$$

$$\text{Hermitian properties: Eigenvalues real, eigenvectors orthogonal, complete (in finite dims).} \quad (10)$$

$$\text{Born rule: } P(\lambda) = |\langle e_\lambda | \Psi \rangle|^2. \quad (11)$$